The Application of Information Theory in Studying Motor Control

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Abstract

Being one of the pillars of the cognitive revolution in psychology, information theory provides valuable insights into how the brain functions as an information system in various cognitive functions. In motor control, the body serves as a conduit for information transmission between the mind and the physical world. With this approach, researchers have demonstrated how optimization of information processing enhances movement efficiency and accuracy, with far-reaching impacts on robotics and AI in terms of replicating human motor learning and control. Overall, the information-theoretic perspective provides a comprehensive evaluation framework that allows disparate cognitive functions to be studied under a unified, information-centric paradigm.

1. Information Theoretic framework: An Integrative Perspective on Cognition

The information-theoretic framework has been embraced across various cognitive domains such as language and motor functions, with research efforts striving to integrate these complex systems within a unified theoretical perspective. This approach, predicated on Shannon's (1948) foundational work in information theory, interprets the brain as a sophisticated information processor, deciphering cognitive processes through the lens of entropy, uncertainty, and predictive computations (Friston, 2010). Such a framework proposes a cohesive mathematical language that links multiple levels of cognitive analysis, from the intricacies of neural circuitry to the overarching algorithms, culminating in an integrated understanding of behavior (Piccinini & Scarantino, 2011). This chapter starts by introducing fundamental concepts of the information-theoretic approach, followed by its application in studies of motor control, and finally an example demonstrating how it unifies the assessment of motor control and higher order cognitive functions. Through this brief introduction, we hope to attract more attention on the application of the information-theoretic approach in profiling individuals' cognitive functions meaningfully and quantitatively from an integrative perspective.

The information theoretic approach refers to analyzing and modeling phenomena using concepts and tools from information theory. The key aspects of the information theoretic approach include:

- Viewing systems and processes in terms of information sources, channels, signals, noise, coding, and communication. This involves representing the system elements mathematically in terms of random variables, probability distributions, entropy, etc.
- Quantifying the amount of information in terms of entropy, mutual information, relative entropy, channel capacity, etc. This allows measuring and comparing the information content of different systems.
- Modeling statistical dependencies and structures in the system using joint and conditional entropies. This captures how the underlying relationships affect the flow of information.
- Looking at optimization of information transmission, such as data compression. Information theory provides fundamental limits and optimal coding schemes.
- Evaluating information loss or distortion due to noise. Measuring the robustness and reliability of communication through the channel.
- Leveraging the mathematical framework of information theory to quantify and analyze the informational aspects of the system or process. This complements physical or mechanistic models.

In its essence, the information theoretic approach provides an abstract, probabilistic framework to study fundamental questions about information and communication properties of systems. It shifts traditional perspectives of various research fields to focus on signals, uncertainties, and statistical relationships transcending the physical details. This approach has been highly influential in fields like telecommunications, data compression, linguistics, and neuroscience. Here, we will use motor control as an example of applying the information theoretic framework in experimental cognitive psychology.

2. Fitts' Law: A New Look at Motor Control through Information Theory

In a groundbreaking paper, Fitts (1954) proposed a computational model based on observations of human performance in a manual aiming task (Fig 1A), positing that the movement time (MT) remains consistent when the ratio between movement amplitude (A) and target width (W) is constant. This implies that longer movements aimed at broader targets take approximately the same amount of time as shorter movements aimed at narrower targets. Furthermore, Fitts observed that the MT escalates when the A to W ratio increases, which can occur through enlarging A, reducing W, or a combination of both. These observations were integrated into a unified equation by Fitts in the following form of Index of Difficulty

(*ID*):

$$ID = \log_2 \frac{2A}{W} \text{ bits}$$

Here *A* stands for the amplitude of the movement, and *W* represents the *precision tolerance*, like the width of the target. This model links the execution of movements to the process of transmitting information. In this framework, movements are quantified by *ID*. The human motor system, in performing a movement task, is conceptualized as transmitting a certain number of "bits of information". By dividing the total bits by the time taken for the movement, one can determine a transmission rate in "bits per second."

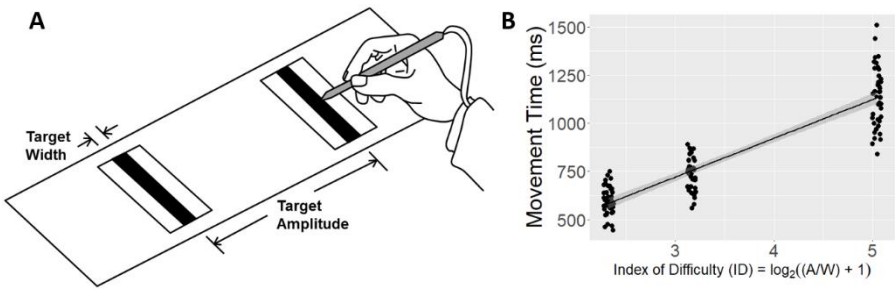

Fig. 1 Fitts' paradigm and typical outcomes. A. Fitts' original experimental settings of the manual tapping task (Fitts, 1954); B. Typical outcomes of movement time as a function of the index of difficulty. Each dot represents the average movement time of a participant under the ID designated on the x-axis.

Following the assessment of information associated with a particular movement, Fitts introduced the Index of Performance (I_p) to measure information capacity by dividing the *ID* by the movement time (MT), which is equivalent to the slope of the linear function between MT and ID (Fig 1B). Fitts' model is conceptually linked to Hick's law in speed-reaction time experiments (Hick, 1952a), which suggests that an increase in the number of alternatives leads to higher uncertainty and entails more information. Due to limited information capacity, processing responses with higher loading of information contents will take more time.

Fitts' original information calculation can result in zero or even negative information values. Specifically, zero information is elicited when the amplitude of movement (*A*) is equal to half the precision tolerance *(W/2)*, while negative information is generated when *W* exceeds *A/2*. In response to these nuances, several modifications of Fitts' law have been proposed to enhance the information provided by aiming movements. For instance, Welford et al. (1969) introduced the concept of the 'landing procedure' for movements approaching the edge of the target position. They suggested that movement terminations should be concentrated in the center of the target, leading to a modified *ID* formula:

$$\log_2 \frac{A + \frac{1}{2}W}{W}$$

Additionally, an analogy with Shannon's Theorem 17 (Shannon, 1948) was incorporated to quantify the *ID* in a different way, as proposed by Mackenzie (1992):

$$\log_2 \frac{A+W}{W}$$

Mackenzie's formulation of *ID* is more appropriate when the signal transmission is subject to increasing noise or decreasing signal quality, whereas Fitts' *ID* is more suitable when a substantial signal is transmitted with low noise. (MacKenzie, 1989). Importantly, Mackenzie's *ID* avoids the generation of negative information when describing relatively high precision tolerance movements. These refinements in ID calculations allow for a more accurate assessment of information capacity in different motor control contexts.

I_p, the *ID* to MT ratio, is a critical metric used to gauge the information processing rate during aiming movements, measured in bits per second. This measure is based on the central assumption that information capacity remains relatively constant across various movement parameters.

However, empirical observations have shown that I_p is not consistent across different *ID* values within the same type of motor task (Zhai, 2004). To further explore the relationship between MT and *ID*, researchers have employed a linear regression model, expressed as $MT = a + b \times ID$. The parameters a and b can be determined by specific experimental motor task. What is particularly noteworthy is that $1/b$ has been found to be more consistent than I_p in quantifying the information processing rate for the same type of movement (Zhai, 2004). This suggests that the inverse of the slope b provides a more stable measure for assessing the rate of information processing in motor tasks, allowing for more accurate comparisons and evaluations across different conditions and contexts.

In an ideal scenario, the y-intercept of Fitts' law should be zero, implying a predicted MT of 0 ms for a task with an ID of 0. This intuitive expectation arises from the notion that a movement with ID = 0 should theoretically require no information to process. However, the persistence of a non-zero y-intercept in Fitts' law posed a challenge from its inception (Fitts & Peterson, 1964; Soukoreff & MacKenzie, 2004). In certain studies, negative y-intercept values have even been observed (Fitts & Peterson, 1964; Kantowitz & Elvers, 1988). Fitts acknowledged the physical impossibility of a zero or negative MT and asserted that the y-intercept should theoretically always be positive (Fitts & Radford, 1966). Various interpretations of non-zero y-intercepts include inherent delays in the psychomotor system (Fitts & Radford, 1966) and uncontrollable muscle activity at the initiation or conclusion of the movement task (Mackenzie, 1992).

Over the years since its initial introduction by Fitts, this law has become one of the most enduring and frequently cited models in experimental psychology. Its applicability and validity have been consistently confirmed through psychomotor studies in various environments, ranging from microscopic to aquatic settings. These studies have repeatedly found strong correlations between the index of difficulty, as per Fitts' model, and the time required to complete a movement task. The fields of kinematics and human factors have been particularly enriched by research utilizing Fitts' analogy to explore aspects of human performance.

3. Factors Modulating Parameters of Fitts' Law

Based on the premise that information capacity varies with specific motor control processes, different motor-related factors are likely to modulate this capacity in the context of Fitts' law. Fitts noted that the characteristics of the movement and tool properties have a minimal impact on information processing (Fitts, 1954). However, Radwin et al. (1990) found that both the direction of movement and the executing effector can influence the Index of Difficulty (ID).

Furthermore, research indicates that movements involving the fingers and wrist over short distances have higher information processing rates (measured in bits/sec) compared to longer arm movements. Langolf et al. (1976) attributed this difference to the engagement of different limb segments. This variation is also explainable through the concept of degrees of freedom in movement, which can increase the steepness of Fitts' law's slope for more distant movements. The complexity of movements, related to degrees of freedom, resonates with Hick's law. Yet, Bertucco et al. (2013) observed no changes in the slope of Fitts' law when manipulating trunk rotation angles. Instead, they noted alterations in the intercept, which might be attributed to Coriolis forces.

Kinetic factors have also been examined in the context of Fitts' law. Loaded movements have been found not modulating the slope of Fitts' law but affecting the intercept (Bonnetblanc, 2008). These kinetic factors can involve various aspects of movement control, such as elbow angle, acceleration/deceleration duration, peak velocity, and EMG responses. However, these kinetic constraints unrelated to the complexity of movement may not affect the information capacity. Also, sensorimotor experience of tolerance levels have been shown to modulate the intercept but not the slope of Fitts' law (Tang et al., 2018). Interestingly, changes in the amplitude do not appear to affect the intercept and slope of the Fitts' law equation. The intercept modulation of kinetic constraints and sensorimotor experience suggest that the "internal" noise level adopted by the subject can be modulated by experience and may further alter the *ID*. On the other hand, the degree of freedom in movement could modulate the information capacity of movement.

4. Dynamic Feedback and Feedforward Information in Online Movement Control

In visuomotor tracking tasks involving visual input (X) and motor output (Y), the use of mutual information to determine the information processing rate is key. This approach includes defining delays and historical depth. Transfer entropy, a critical tool in this context, models the information processed through conditional mutual information, balancing recent output with previous inputs, given the output's history. Lam and Zénon (2021) highlighted the pivotal role of predictability in influencing the information processing rates of feedback and feedforward systems, noting that these effects are inversely related. This insight underscores the necessity for future research to not only quantify motor control information capacity but also dissect the distinct contributions of feedback and feedforward information over time. Such studies should take into account the cumulative impact of visuomotor delay, moving beyond a mere aggregation of motor planning information.

5. Sensory Information Before and During the Movement

The loci of Fitts' law has intrigued researchers: Does its focus lie in initial information acquisition and transmission or in online control aspects? Wu et al. (2010) separated motor control into stages like cognition, planning, feedback, and control. They showed visual feedback impacts the slope relating MT and ID. Limiting feedback increased MT variations and reduced accuracy, highlighting visual feedback's role. While valuable, Fitts' law may not fully capture online control complexity. In visuomotor tracking tasks, for example, mutual information is key, encompassing delays and history. The finite capacity of human control, bounded by noise, implies an upper limit on processing rate. This relates to the speed-accuracy trade-off, where faster movements enable lower precision.

While Fitts' Law provides a valuable perspective on the initial stages of information processing, it may not fully capture the complex mechanisms of online control. For example, in visuomotor tracking tasks, where visual input (X) aids in motor output (Y), mutual information is crucial for quantifying the information processing rate, including defined delays and history depth. The human visuomotor control

system's finite capacity, bounded by noise, mirrors an upper limit in information processing rate. This finite capacity relates to the observed speed-accuracy trade-off in motor tasks, where increased speed (with higher noise tolerance) results in less precision, and vice versa.

Gori et al. (2018) advocated for clearer definitions of capacity and throughput in Fitts' law contexts, enhancing previous interpretations and incorporating concepts like misses at rate ε. This model focuses on minimizing movement time rather than average performance (Gori et al., 2017). Capacity is defined as the maximum mutual information $I(X;Y)$ transmitted through the channel, ensuring reliable data transmission. *Throughput*, the effective data transmission speed, is defined as MT_i / ID_i in motor tasks, representing effective information transmission speed. Soukoreff and MacKenzie (2009) introduced the concept of *equivocation*, namely the difference between throughput and source entropy, to model the attainable region on source entropy to equivocation plots. The source entropy processing range, limited by channel capacity, expands with increased precision tolerance, aligning with the speed-accuracy trade-off.

Previous studies focused on the maximum capacity constrained to the Eh line in source entropy to *equivocation plots*. Soukoreff and MacKenzie (2009) noted human motor performance often deviates from ideal information transmission lines, introducing a constant (α) to modulate the human-achievable equivocation: Eh = α(I-C). This approach, using the best MT to infer Fitts' law, takes into account field study variations in MT. Gori et al. (2017) suggested to distinguish between *deviation of movement termination* (spatial error) and *misses* (*binary hit-or-miss indicator*, i.e., *erasure*), and proposed a new effective index ID(ε): $(1-\varepsilon)\log_2\frac{A+W}{W}$, where ε indicates the target miss rate. The intricate measurements and integration of *source entropy*, *information capacity*, *throughput*, and *equivocation* enriches the understanding of the nature of human motor performance.

6. Bridging Motor Skills and Cognition: An Information-Theoretic Nexus

While information theory is foundational to the "cognitive revolution" as highlighted by Neisser in 1976, its application in cognitive psychology has often been more metaphorical than quantitative. Typically, cognitive psychology experiments do not directly quantify the amount of information in stimuli in relation to cognitive processing. However, an information theory-based approach can provide a straightforward and precise means of measuring cognitive functions. Take, for instance, the assessment of executive function using tasks that involve processing uncertainty, such the Majority Function Task (MFT; Fan, 2014; Fan et al., 2008; Fan et al., 2014; Wang et al., 2011). In MFT, participants are presented with arrays of arrows pointing left or right and must identify the predominant direction. The task's complexity is varied by adjusting the number of arrows and their arrangement (congruency). The researchers hypothesize different search strategies like exhaustive, self-terminating, or grouping searches. This task's computational demand for each trial is quantified in bits of entropy. Fan et al. (2008) found that the grouping search strategy most effectively represented the linear relationship between reaction time (RT) and computational load in MFT, where RT is expressed as a function of the average number of arrows scanned per trial (s): $\mathrm{RT} = \mathrm{a} + \mathrm{b}\times\log2(\mathrm{s})$. Analogous to the Fitts' law, here the slope (**b**) represents the additional processing time required per bit of increased load, indicating processing efficiency. On the other hand, the intercept (**a**) corresponds to the processing time for the simplest task possible (0 bit), akin to the reaction time for responding to a single arrow. Thus, information theory provides a precise means to measure cognition in uncertain tasks like MFT.

The parallels between the MFT and Fitts' Law are evident, as both exhibit a linear relationship between a measure of speed and the quantity of information processed. This analogy extends to a comparison made by Sleimen-Malkoun et al. (2013) between Fitts' law and Hick-Hyman's law in young and elderly groups. Hick-Hyman's Law, in contrast to Fitts' Law, characterizes the choice reaction time (CRT) as a linear function of the information entropy in response selection, essentially the binary logarithm of potential stimulus-response associations, while maintaining the motor response complexity at its minimum (Hawkins et al., 2012; Hick, 1952b; Hülür et al., 2015; Hyman, 1953). The study revealed that, although

the slopes of both laws were indistinguishable in the elderly, younger participants exhibited steeper slopes in Fitts' Law compared to Hick-Hyman's Law.

Following the same vein, Chang (2021) compared aging across various cognitive domains using experimental designs that manipulate computational load within a consistent informational framework. To quantitatively assess these designs, linear functions for the MFT and Fitts' Law were evaluated using the Generalized Linear Mixed Model (GLMM), which examined the interplay of age, task, and computational load both at a collective level (as a fixed effect) and on an individual basis (as a random effect). The outcomes indicated that among the elderly, increasing computational load led to similar rates of increment in movement time (MT) and reaction time (RT). In contrast, younger participants exhibited a steeper RT slope in the MFT compared to the MT slope in Fitts' Law (**Fig. 2**). Notably, stronger correlations were observed in the elderly group between individual slope and intercept estimates across and within tasks, compared to the younger group. In addition, the observed stronger cross-domain correlations further corroborate the dedifferentiation hypothesis. This research approach facilitates more direct cross-domain comparisons than would be possible with different tasks that do not align in computational load. Without this alignment, comparisons would rely on summarized and transformed data, potentially reducing statistical power and limiting inferences about lifespan trajectories across cognitive domains.

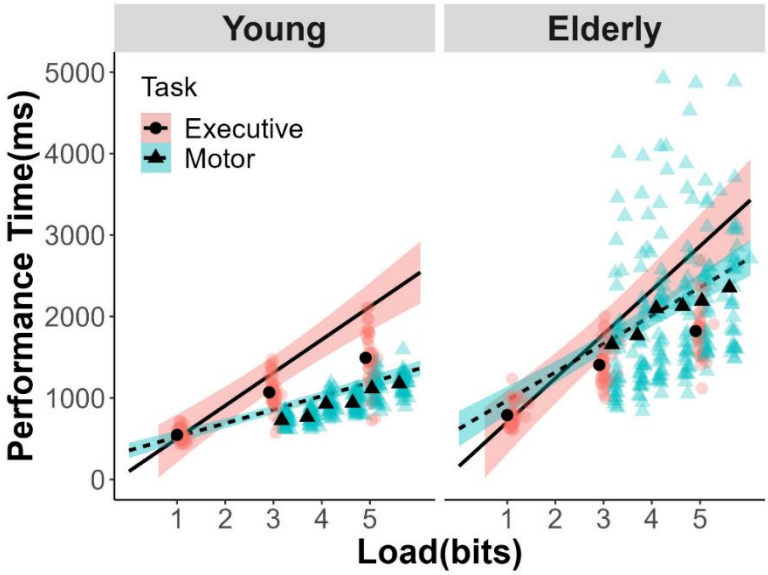

Fig. 2 Predicted values of performance time as a function of Age Group and Task. The solid line (pinkish shade) and dashed line (cyan shade) indicate the linear function (95% CI) between performance time and Load in the MFT (executive) and Fitts (motor) task, respectively. The dot and triangle symbols represent the group mean of the analysed data at each combination between Task and Load in the experiment, with error bars indicate the 95% CI normalized for within-subject error (adapted from Chang [2021]).

Synthesizing the findings from studies that compare motor and cognitive functions within the information-theoretic framework, there is support for the cognitive aging dedifferentiation hypothesis. This hypothesis posits that with age-related reductions in cognitive resources, distinct cognitive processes that are separated in youth tend to converge and use shared resources in older age, leading to increased similarity (Baltes & Lindenberger, 1997; Lindenberger & Baltes, 1994). However, it's important to note that the evidence supporting the dedifferentiation hypothesis has been mixed (Hülür et al., 2015), potentially influenced by research design (longitudinal vs. cross-sectional) and the nature of the sample (including life-span cohorts with young and middle-aged participants or focusing solely on the elderly).

Besides these factors, a third potential influence arises from the application of distinct methods used to measure and analyze different cognitive functions, which might not be comparable and can significantly affect the results and interpretations of aging trends. La Fleur et al. (2018) illustrated this point by showing that including a speed of processing test in correlation analyses with other abilities influenced the

tendency for linear dedifferentiation among cognitive domains. Since studies on dedifferentiation typically rely on distinct methods to quantify various cognitive domains, the resultant analyses and interpretations are often based on transformed scores. Thus, considering the current study's findings, along with those from previous reports, it appears that some degree of variation in the inconsistencies across studies might be attributable to distortions in measurement resulting from the transformation and averaging of performance indices. To this end, designing cognitive task and measuring performance under the information theoretic framework may provide a fair perspective for cross-domain comparison.

7. Other Applications of the Information Theoretic Approach

It is somewhat paradoxical that core information metrics are often absent in cognitive psychology experiments, a field that professes to be grounded in information processing theory. The adoption of information-entropy-based manipulations across various cognitive abilities might necessitate a significant redesign of tasks that are currently structured around widely accepted operational definitions. Fan (2014) has demonstrated how to reframe existing cognitive control tasks within the information theoretic paradigm. We can interpret conflict effects stemming from an interfering dimension as examples of variations in uncertainty between conflict and non-conflict conditions. In the Stroop task, for instance, the conflict arises from the discrepancy between the color of the text (one dimension) and the word meaning (another dimension). This conflict, manifesting as a discrepancy between the color of the text and the word meaning, increases the entropy in the system. Entropy, in this context, refers to the level of uncertainty or disorder in processing information. When the brain encounters a word like "RED" printed in blue ink, it faces a higher entropy situation due to the incongruence between the color and the word meaning. Resolving this incongruence requires the brain to suppress the automatic reading response (a lower entropy task) in favor of the more complex task of naming the ink color (a higher entropy task). This increased cognitive effort due to higher entropy is reflected in slower response times and a greater likelihood of errors, illustrating the impact of conflicting information dimensions in cognitive processing.

Another good example is the flanker task. Through an information-theoretic lens, the conflict in the flanker task arises from the presence of distracting flankers around a central target introduces increased entropy, signifying higher uncertainty in the cognitive process. When participants encounter incongruent flankers, the brain has to manage conflicting streams of information – the direction of the central arrow (target) and the surrounding arrows (flankers). This situation heightens the cognitive load due to the need to suppress responses to the misleading flanker information. Consequently, the increased entropy manifests as slower reaction times and a higher error rate in incongruent conditions, compared to congruent ones where flanker and target information align, resulting in lower entropy and easier cognitive processing.

Moreover, one could quantify uncertainty using surprise and entropy calculations in paradigms that investigate phenomena like the oddball effect, Go/No-Go tasks, or task switching, which all involve contrasting responses to events of varying probabilities. For example, in the task-switching paradigm, the cognitive challenge arises from the need to switch between different tasks, each with its own set of rules and responses. This switch introduces high entropy – an increase in uncertainty and cognitive load – as the brain must inhibit the rules and responses of the previous task while activating those of the new task. This increased entropy is evidenced by the typically observed switch cost, manifested as slower response times and higher error rates when switching tasks compared to repeating the same task. This reflects the brain's effort to reconfigure and stabilize a new cognitive state, aligning with principles of information theory.

The information theoretic paradigm also shows promise for elucidating mechanisms of language processing. For instance, experiments could manipulate linguistic uncertainty by varying lexical frequency and syntactical complexity in sentences and passages (Hale, 2001). Increased surprisal and entropy from rare words and convoluted syntax require heightened cognitive effort, measurable via slower reading times, more errors in comprehension questions, and modulated neurological signals (Levy, 2008). Moreover, analyzing linguistic inputs and outputs in terms of transmitted bits of information per unit time

provides a quantitative basis to compare efficiency of language processing across tasks and populations (Jaeger, 2010). An information-centric framework centered on quantifying transmission, storage and loss of linguistic information at multiple levels ranging from phonemes to semantics could yield fresh insights into the nature of the language system(Linzen & Jaeger, 2015). Going further, some researchers envision the development of a "calculus of thought" founded on information theoretic formulations, providing an integrated perspective on cognition encompassing perception, executive functions, memory, language and more under a unified set of information processing principles (Piccinini & Scarantino, 2011). Applying this approach to complex domains like long-term memory and language is challenging initially. However, as the information framework expands to encompass basic cognitive processes, it may evolve into a comprehensive system that explicates cognitive functions regarding information exchange and storage (Wolff, 1982). This unified analysis under common information metrics could transform understanding of cognition across domains (Dell, 1986). Indeed, the information paradigm heralds new horizons for cognitive science, propelling more precise and integrated models of the human mind.

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